MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, ADVANCED SCHOTTKY TTL, 4-BIT BINARY FULL ADDERS WITH FAST CARRY, MONOLITHIC SILICON

This specification is approved for use by the Rome Air Development Center, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, Advanced Schottky TTL, 4-Bit Binary Full Adder microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, and as specified herein.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	4-Bit Binary Full Adder (with fast carry)

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
E	D-2 (16-lead, 1/4" x 7/8", dual in-line package)
F	F-5 (16-lead, 1/4" x 3/8", flat package)
2	C-2 (20-terminal, .350" x .350"), square chip carrier package

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V dc to +7.0 V dc
Input voltage range	-1.2 V dc at -18 mA to +7.0 V dc
Storage temperature range	-65°C to +150°C
Maximum power dissipation (P_D) 1/	302 mW
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction-to-case (θ_{JC}):	
Case E	50°C/W
Case F	70°C/W
Case 2	60°C/W
Junction temperature (T_J)	+175°C

1/ Must withstand the added P_D due to short circuit test (e.g., I_{OS}).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Recommended operating conditions.

Supply voltage	4.5 V dc minimum to 5.5 V dc maximum
Minimum high-level input voltage (V_{IH})	2.0 V dc
Maximum low-level input voltage (V_{IL})	0.8 V dc
Normalized fanout (each output) 2/	
Low logic level	16 maximum
High logic level	50 maximum
Case operating temperature range (T_C)	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.2 Truth table. The truth table shall be as specified on figure 2.

3.2.3 Logic diagram. The logic diagram shall be as specified on figure 3.

3.2.4 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained and available upon request.

3.2.5 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 and 6.4 herein.

2/ The device shall fanout in both high and low levels to the specified number of inputs of the same device type as that being tested.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \leq T_C \leq +125°C$	Limits		Unit
			Min	Max	
High-level output voltage	V_{OH}	V_{CC} = 4.5 V, I_{OH} = -1.0 mA, I_{IL} = 0.8 V, V_{IH} = 2.0 V	2.5		V
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V, I_{OL} = 20 mA, I_{IL} = 0.8 V, V_{IH} = 2.0 V		0.5	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -18 mA		-1.2	V
High-level input current	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.7 V		20	µA
	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 7.0 V		100	µA
Low-level input current	I_{IL1}	V_{CC} = 5.5 V, V_{IN} = 0.5 V	-0.5	-1.2	mA
	I_{IL2}		-0.25	-0.60	mA
Short-circuit output current 1/	I_{OS}	V_{CC} = 5.5 V	-60	-150	mA
Output Drive	I_{OD}	V_{CC} = 4.5 V, V_{IN} = 5.5 V	60		mA
Supply current	I_{CC1}	V_{CC} = 5.5 V		55	mA
	I_{CC2}			55	mA
	I_{CC3}			55	mA
Propagation delay time, low to high level	t_{PLH1}	V_{CC} = 5.0 V, C_L = 50 pF, ±10% R_L = 500Ω ±5%	1.0	14.0	ns
	t_{PLH2}		1.0	10.5	ns
	t_{PLH3}		1.0	14.0	ns
	t_{PLH4}		1.0	10.5	ns
Propagation delay time, high to low level	t_{PHL1}		1.0	14.0	ns
	t_{PHL2}		1.0	10.0	ns
	t_{PHL3}		1.0	14.0	ns
	t_{PHL4}		1.0	10	ns

1/ Not more than one output should be shorted at a time.

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3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the microcircuit, but shall be retained on the initial container.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 11 (see MIL-M-38510, appendix E).

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (pre burn-in) (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3,7,9,10,11	1*,2,3,7,9
Group A test requirements (method 5005)	1,2,3,7,8,9,10,11	1,2,3,7,8,9,10,11
Group B test requirements (method 5005) subgroup 5	1,2,3,7,8,9,10,11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1,2,3
Additional electrical subgroups for group C periodic inspections	N/A	N/A
Group D end-point electrical parameters (method 5005)	1, 2, 3	1, 2, 3

*PDA applies to subgroup 1 (see 4.2c).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in (method 1015 of MIL-STD-883).

(1) Test condition D, using the circuit shown on figure 3, or equivalent.

(2) T_A = +125°C minimum.

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b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Electrical test requirements shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical test requirements shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D, using the circuit shown on figure 3, or equivalent.

(2) T_A = +125°C minimum.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspections. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the reference terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

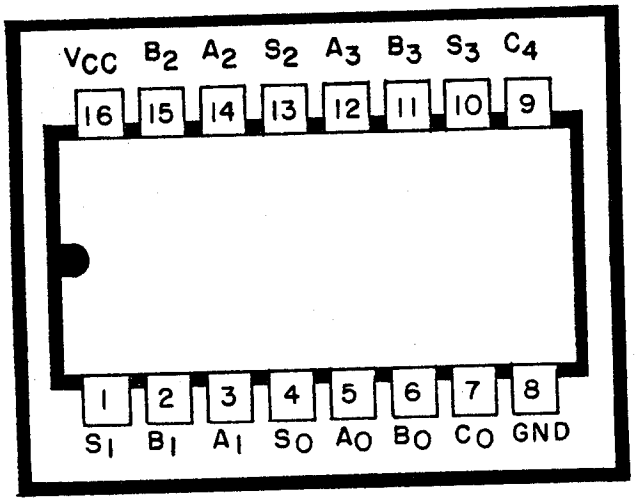
6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

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Device type 01
Cases E and F



Device type 01
Case 2

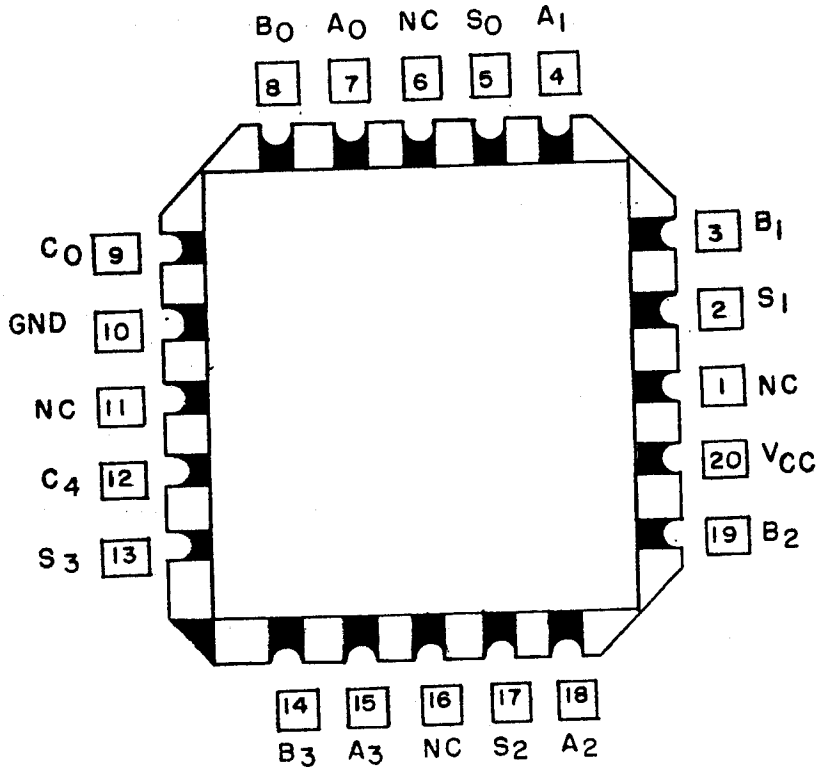


FIGURE 1. Terminal connections (top view).

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Device type 01

INPUT						OUTPUT					
						When C_0 = L			When C_0 = H		
						When C_2 = L			When C_2 = H		
A_0 / A_2	B_0 / B_2	A_1 / A_3	B_1 / B_3	S_0 / S_2	S_1 / S_3	C_2 / C_4	S_0 / S_2	S_1 / S_3	C_2 / C_4		
L	L	L	L	L	L	L	H	L	L		
H	L	L	L	H	L	L	L	H	L		
L	H	L	L	H	L	L	L	H	L		
H	H	L	L	L	L	L	H	L	L		
L	L	H	L	L	H	L	H	L	L		
H	L	H	L	H	H	L	L	H	L		
L	H	H	L	H	H	L	L	H	L		
H	H	H	L	L	H	L	H	H	L		
L	L	L	H	H	L	L	H	L	H		
H	L	L	H	L	L	L	L	H	H		
L	H	L	H	L	L	L	L	H	H		
H	H	L	H	H	L	L	H	L	H		
L	L	H	H	H	H	L	L	H	H		
H	L	H	H	L	H	H	H	H	H		
L	H	H	H	L	H	H	L	L	H		
H	H	H	H	H	H	H	H	H	H		

NOTE: Input conditions at A_0, B_0, A_1, B_1 and C_0 are used to determine outputs S_0 and S_1 and the value of the internal carry C_2. The values at C_2, A_2, B_2, A_3 and B_3 are then used to determine outputs S_2, S_3 and C_4.

FIGURE 2. Truth table.

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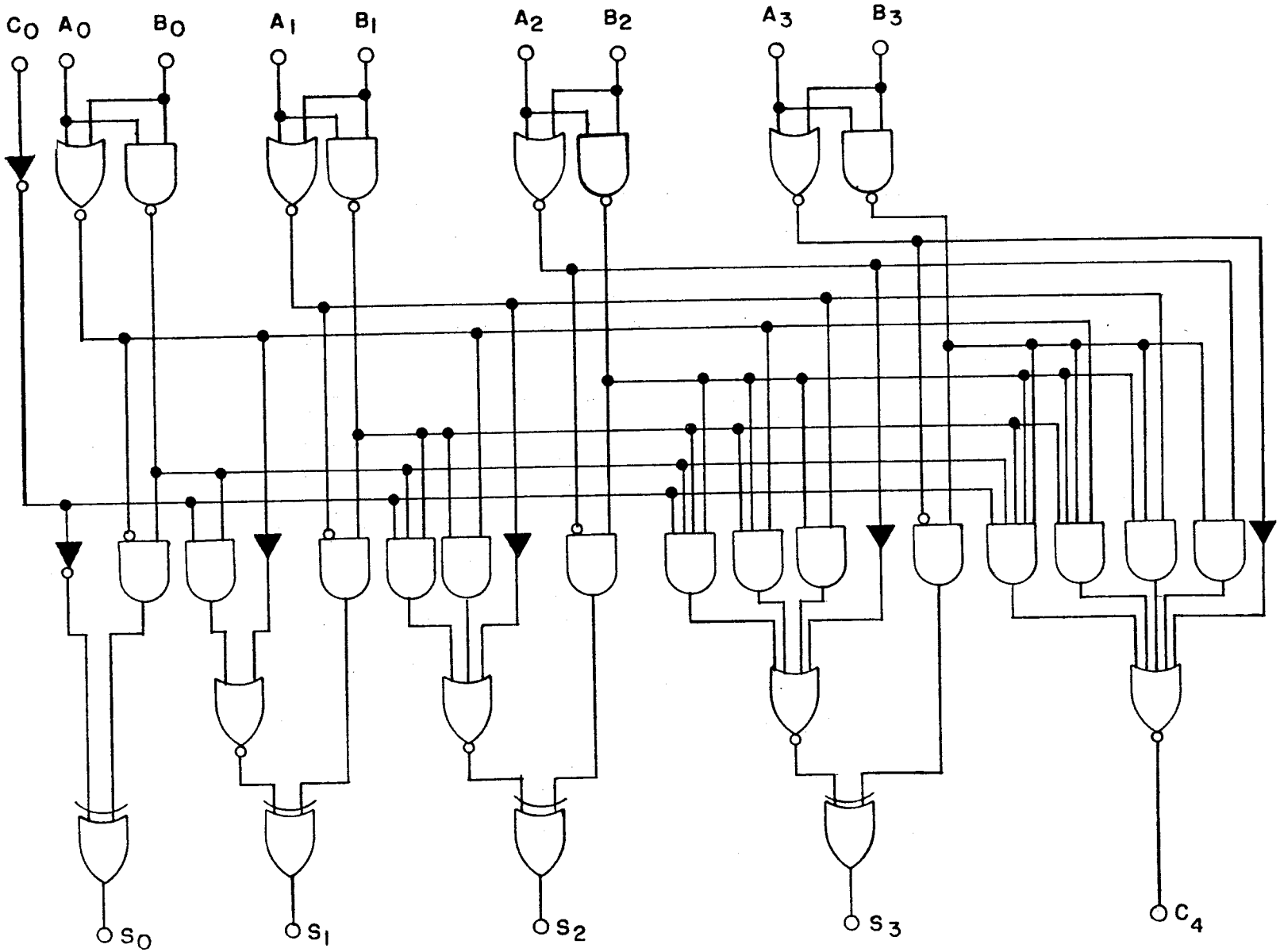
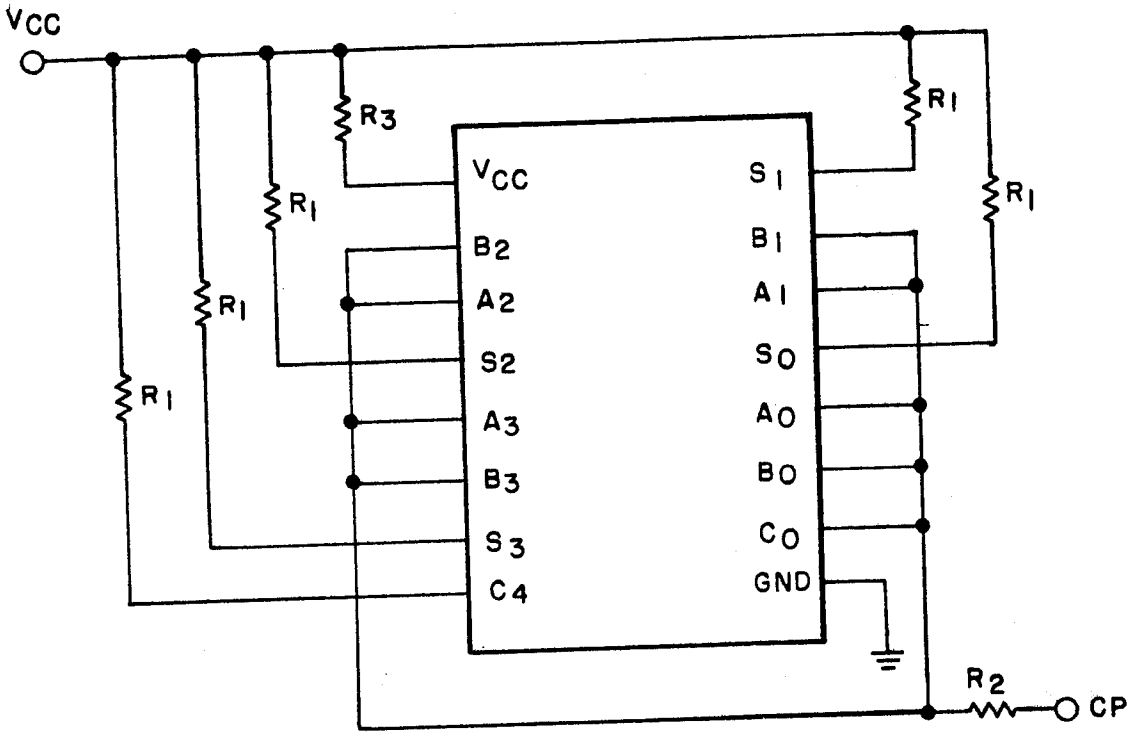


FIGURE 3. Logic diagrams - Continued.

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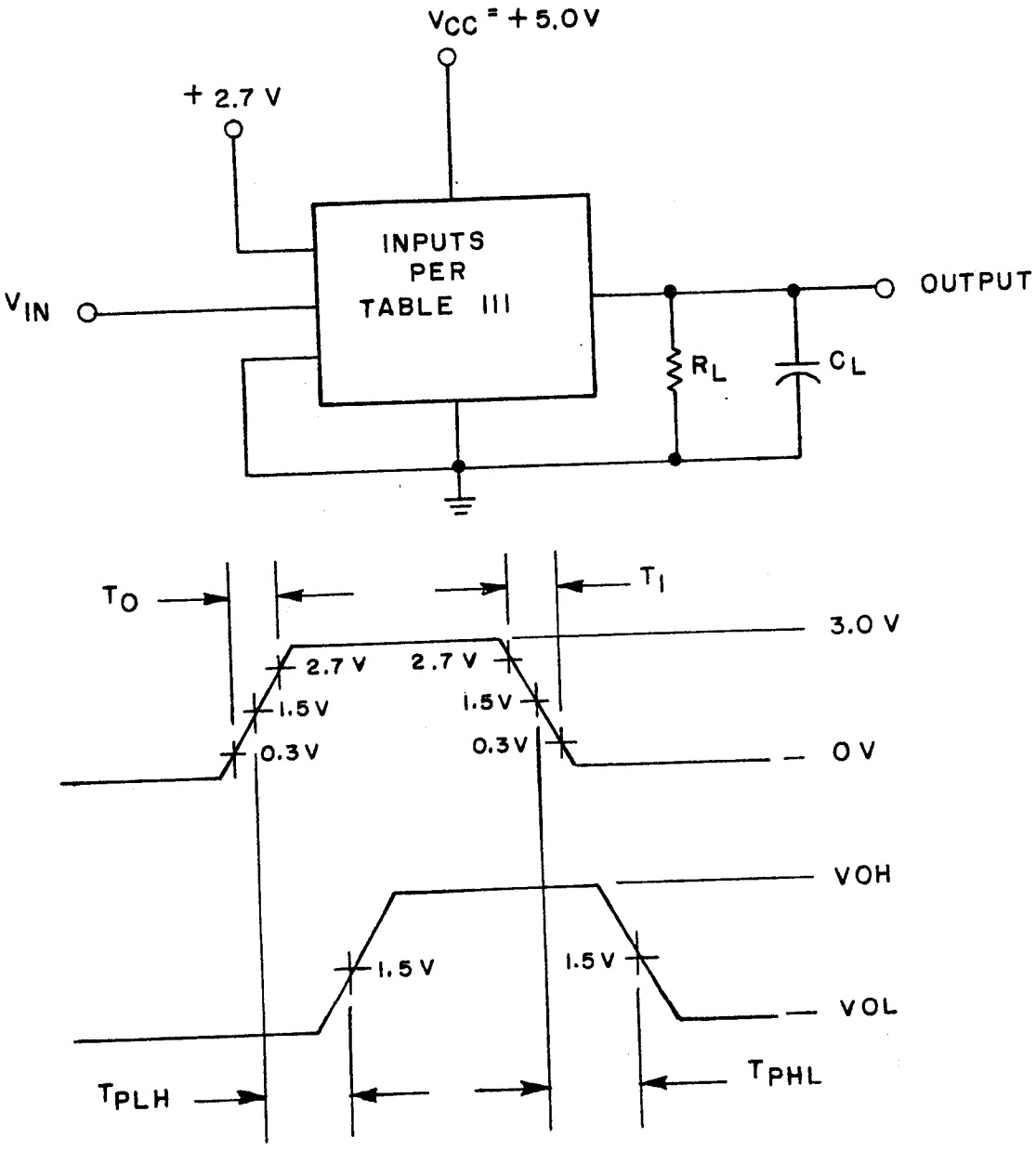
Device type 01



NOTES:
1. CP = 100 kHz ±50%; duty cycle = 50 ±15%; V_{IH} = 2.0 V minimum to 5.5 V maximum; V_{IL} = -0.5 V minimum to 0.8 V maximum.
2. R1 = 260Ω ±5%; R2 = 51Ω ±5%.
3. V_{CC} and R3 shall be chosen to insure a 5.5 V minimum at device V_{CC} terminal.

FIGURE 4. Burn-in and steady state life test circuit.

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NOTES:
1. $T_0 = T_1 \leq 2.5$ ns; PRR ≤ 1 MHz, $Z_{OUT} \approx 50\Omega$.
2. $C_L = 50$ pF ±10%, including scope probe, wiring, and stray capacitance without package in test fixture.
3. $R_L = 499\Omega$ ±5%.
4. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 5. Switching time waveforms for device type 01.

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TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Case 1/ 2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
				2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	S1	B1	A1	S0	A0	B0	C0	GND	C4	S3	B3	A3	S2	A2	B2	V_{CC}				
1 T_C = +25°C	V_{OH}	3006	1	-1 mA	2.0 V	2.0 V	-1 mA	2.0 V	2.0 V	2.0 V	GND	-1 mA	-1 mA	2.0 V	2.0 V	-1 mA	2.0 V	2.0 V	4.5 V	S0	2.5		V
		"	2		"	"		"	"	"	"			"	"		"	"	"	S1	"		"
		"	3		"	"		"	"	"	"			"	"		"	"	"	S2	"		"
		"	4		"	"		"	"	"	"			"	"		"	"	"	S3	"		"
		"	5		"	"		"	"	"	"			"	"		"	"	"	C4	"		"
	V_{OL}	3007	6	20 mA	0.8 V	0.8 V	20 mA	0.8 V	0.8 V	0.8 V	"	20 mA	20 mA	0.8 V	0.8 V	20 mA	0.8 V	0.8 V	"	S0		0.5	"
		"	7		"	"		"	"	"	"			"	"		"	"	"	S1		"	"
		"	8		"	"		"	"	"	"			"	"		"	"	"	S2		"	"
		"	9		"	"		"	"	"	"			"	"		"	"	"	S3		"	"
		"	10		2.0 V	2.0 V		2.0 V	2.0 V	"	"			"	"		"	"	"	C4		"	"
	V_{IC}		11		-18 mA						"								"	B1		-1.2	"
			12			-18 mA					"								"	A1		"	"
			13					-18 mA			"								"	A0		"	"
			14						-18 mA		"								"	B0		"	"
			15							-18 mA	"								"	C0		"	"
			16								"			-18 mA					"	B3		"	"
			17								"				-18 mA				"	A3		"	"
			18								"						-18 mA		"	A2		"	"
			19								"							-18 mA	"	B2		"	"
	I_{IH1}	3010	20		GND	2.7 V					"								5.5 V	B1		20	μA
		"	21		2.7 V	GND					"								"	A1		"	"
		"	22					2.7 V	GND		"								"	A0		"	"
		"	23					GND	2.7 V		"								"	B0		"	"
		"	24								"			GND	2.7 V				"	A3		"	"
		"	25								"			2.7 V	GND				"	B3		"	"
		"	26								"						2.7 V	GND	"	A2		"	"
		"	27								"						GND	2.7 V	"	B2		"	"
		"	28							2.7 V	"								"	C0		"	"
	I_{IH2}	"	29		GND	7.0 V	7.0 V				"								"	A1		100	"
		"	30		7.0 V	GND					"								"	B1		"	"
		"	31					7.0 V	GND		"								"	A0		"	"
		"	32					GND	7.0 V		"								"	B0		"	"
		"	33								"			GND	7.0 V				"	A3		"	"
		"	34								"			7.0 V	GND				"	B3		"	"
		"	35								"						7.0 V	GND	"	A2		"	"
		"	36								"						GND	7.0 V	"	B2		"	"
		"	37							7.0 V	"								"	C0		"	"
	I_{IL1}	3009	38		5.5 V	0.5 V					"								"	A1	2/	2/	mA
		"	39		0.5 V	5.5 V					"								"	B1	"	"	"
		"	40					0.5 V	5.5 V		"								"	A0	"	"	"
		"	41					5.5 V	0.5 V		"								"	B0	"	"	"
		"	42								"			5.5 V	0.5 V				"	A3	"	"	"
		"	43								"			0.5 V	5.5 V				"	B3	"	"	"
		"	44								"						0.5 V	5.5 V	"	A2	"	"	"
		"	45								"						5.5 V	0.5 V	"	B2	"	"	"
	I_{IL2}	"	46							0.5 V	"								"	C0	"	"	"

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Case 1/ 2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	S1	B1	A1	S0	A0	B0	C0	GND	C4	S3	B3	A3	S2	A2	B2	V_{CC}				
1 T_C = +25˚C	I_{OS}	3011	47	GND	5.5 V	5.5 V	GND	5.5 V	5.5 V	5.5 V	GND	GND	GND	5.5 V	5.5 V	GND	5.5 V	5.5 V	4.5 V	S1 S0 S3 S2 C4	-60	-150	mA
			48		"	"		"	"	"	"			"	"		"	"	"		"	"	"
			49		"	"		"	"	"	"			"	"		"	"	"		"	"	"
			50		"	"		"	"	"	"			"	"		"	"	"		"	"	"
			51		"	"		"	"	"	"	GND		"	"		"	"	"		"	"	"
	I_{OD}		52	2.5 V	GND	GND	2.5 V	GND	GND	GND	"		2.5 V	GND	GND	2.5 V	GND	GND	"	S1 S0 S3 S2 C4	60		"
			53							"	"								"		"		"
			54							"	"								"		"		"
			55							"	"								"		"		"
			56		5.5 V	5.5 V		5.5 V	5.5 V	"	"	2.5 V		GND	GND		5.5 V	GND	"		"		"
	I_{CC1}	3005	57		GND	GND		GND	GND	"	"			GND	GND		GND	GND	5.5 V	V_{CC}		55	"
	I_{CC2}	"	58		GND	5.5 V		5.5 V	GND	5.5 V	"			GND	5.5 V		5.5 V	GND	5.5 V	"		55	"
	I_{CC3}	"	59		5.5 V	5.5 V		5.5 V	5.5 V	5.5 V	"			5.5 V	5.5 V		5.5 V	5.5 V	5.5 V	"		55	"
2	Same tests, terminal condition, and limits as for subgroup 1, except T_C = +125˚C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55˚C and V_{IC} tests are omitted.																						
7 T_C = +25˚C 3/	Truth table test	3014	60	B	L	L	B	L	L	L	GND	B	B	L	L	B	L	L	5.0 V	All Outputs	See 4/		
			61	"	L	L	A	L	L	H	"	A	A	H	H	B	H	H	"				
			62	"	L	L	B	L	L	L	"	A	B	H	H	A	L	H	"				
			63	A	L	L	A	L	H	H	"	A	A	H	H	B	H	L	"				
			64	B	L	L	B	L	H	L	"	A	B	H	H	A	L	L	"				
			65	A	L	L	B	L	H	H	"	B	A	H	H	A	H	L	"				
			66	"	L	L	B	L	H	L	"	B	A	H	H	B	L	H	"				
			67	"	L	L	B	L	L	H	"	B	B	H	L	A	H	H	"				
			68	A	L	H	B	L	L	L	"	B	A	H	L	B	H	H	"				
			69	A	L	H	A	L	L	H	"	B	A	H	L	A	H	L	"				
			70	A	L	H	B	L	L	L	"	B	A	H	L	B	H	L	"				
			71	B	L	H	A	L	H	H	"	B	A	H	L	A	H	L	"				
			72	A	L	H	B	L	H	L	"	B	B	H	L	A	H	L	"				
			73	B	L	H	A	L	H	H	"	B	B	H	L	B	H	L	"				
			74	B	L	H	B	L	H	L	"	B	B	H	L	B	H	L	"				
			75	B	L	H	B	L	L	H	"	B	B	H	L	B	H	L	"				
			76	A	H	L	B	L	L	L	"	B	A	H	L	B	H	L	"				
			77	A	H	L	A	L	L	H	"	B	A	H	L	A	H	L	"				
			78	A	H	L	B	L	L	L	"	B	A	H	L	B	H	L	"				
			79	B	H	L	A	L	H	H	"	B	A	H	L	A	H	L	"				
			80	A	H	L	B	L	H	L	"	B	B	H	L	A	H	L	"				
			81	B	H	L	A	L	H	H	"	B	B	H	L	B	H	L	"				
			82	B	H	L	B	L	H	L	"	B	B	H	L	B	H	L	"				
			83	B	H	L	B	L	L	H	"	B	A	H	L	A	H	L	"				
			84	B	H	H	B	L	L	L	"	A	B	H	L	B	L	H	"				
			85	B	H	H	A	L	L	H	"	A	B	H	L	A	L	H	"				
			86	B	H	H	B	L	H	H	"	A	A	H	L	B	L	H	"				
			87	A	H	H	A	L	H	L	"	A	A	H	L	A	L	H	"				
			88	B	H	H	B	L	H	L	"	A	A	H	L	B	L	H	"				
			89	A	H	H	A	L	H	H	"	A	A	H	L	A	L	H	"				
			90	A	H	H	B	L	H	L	"	A	B	H	L	B	L	H	"				
			91	A	H	H	A	L	H	H	"	B	B	H	L	A	L	H	"				
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = 125˚C and T_C = 55˚C.																						

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Case 1/ 2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits Min	Limits Max	Unit
				2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Test no.	S1	B1	A1	S0	A0	B0	C0	GND	C4	S3	B3	A3	S2	A2	B2	V_{CC}				
9 T_C = +25°C	t_{PLH1}	3003	92		GND	GND	OUT	2.7 V	GND	IN	GND								5.0 V	CO to S0	1.5	9.5	ns
	t_{PLH1}	Fig. 5	93	OUT	"	2.7 V		"	"	"	"					OUT			"	CO to S1	"	"	"
	t_{PLH1}		94		"	"		"	"	"	"						2.7 V	GND	"	CO to S2	"	"	"
	t_{PLH1}	"	95		"	"		"	"	"	"		OUT	GND	2.7 V		2.7 V	GND	"	CO to S3	"	"	"
	t_{PLH2}	"	96		"	"		"	"	"	"	OUT		GND	2.7 V		2.7 V	GND	"	CO to S4	"	7.5	"
	t_{PLH3}	"	97	OUT	"	IN		GND	"	GND	"								"	A1 to S1	"	9.5	"
	t_{PLH3}	"	98	OUT	IN	GND		GND	"	GND	"								"	B1 to S1	"	"	"
	t_{PLH3}	"	99								"		OUT	GND	IN		GND	GND	"	A3 to S3	"	"	"
	t_{PLH3}	"	100								"		OUT	IN	GND		GND	GND	"	B3 to S3	"	"	"
	t_{PHL1}	"	101		GND	GND	OUT	2.7 V	GND	IN	"								"	CO to S0	"	"	"
	t_{PHL1}	"	102	OUT	"	2.7 V		"	"	"	"					OUT			"	CO to S1	"	"	"
	t_{PHL1}	"	103		"	"		"	"	"	"						2.7 V	GND	"	CO to S2	"	"	"
	t_{PHL1}	"	104		"	"		"	"	"	"		OUT	GND	2.7 V		2.7 V	GND	"	CO to S3	"	"	"
	t_{PHL2}	"	105		"	"		"	"	"	"	OUT		GND	2.7 V		2.7 V	GND	"	CO to S4	"	7.0	"
	t_{PHL3}	"	106	OUT	"	IN		GND	"	GND	"								"	A1 to S1	"	9.5	"
	t_{PHL3}	"	107	OUT	IN	GND		GND	GND	GND	"								"	B1 to S1	"	"	"
	t_{PHL3}	"	108								"		OUT	GND	IN		GND	GND	"	A3 to S3	"	"	"
	t_{PHL3}	"	109								"		OUT	IN	GND		GND	GND	"	B3 to S3	"	"	"
	t_{PHL4}	"	110		IN				2.7 V	2.7 V	"	OUT		2.7 V				2.7 V	"	B2 to C4	"	7.5	"
	t_{PHL4}	"	111		2.7 V	IN			2.7 V	2.7 V	"	"			2.7 V		2.7 V		"	A2 to C4	"	"	"
	t_{PHL4}	"	112		2.7 V						"	"		IN			2.7 V	2.7 V	"	B4 to C4	"	"	"
	t_{PHL4}	"	113								"	"		2.7 V	IN				"	A4 to C4	"	"	"
	t_{PHL4}	"	114		IN				2.7 V	2.7 V	"	"		2.7 V				2.7 V	"	B2 to C4	"	7.0	"
	t_{PHL4}	"	115		2.7 V	IN			2.7 V	2.7 V	"	"			2.7 V		2.7 V		"	A2 to C4	"	"	"
	t_{PHL4}	"	116		2.7 V						"	"		IN			2.7 V	2.7 V	"	B4 to C4	"	"	"
	t_{PHL4}	"	117								"	"		2.7 V	IN				"	A4 to C4	"	"	"
10	t_{PLH1} t_{PLH2} t_{PLH3} t_{PHL1} t_{PHL2} t_{PHL3} t_{PLH4} t_{PHL4}	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C and limits are as shown.																			1.0	14.0	"
																					"	10.5	"
																					"	14.0	"
																					"	14.0	"
																					"	10.0	"
																					"	14.0	"
																					"	10.5	"
																					"	10.0	"
11	Same tests, terminal conditions and limits, as for subgroup 10 except T_C = -55°C.																						

1/ Case 2 pins not referenced are N/C.

2/ Test limits are as follows:

Symbol	Min/Max limits (mA) for circuit: A	B	C
I_{IL1}	-0.50/-1.2		
I_{IL2}	-.25/-.60		

3/ Test numbers 60 through 91 shall be run in sequence.

4/ Inputs A = 3.0 V minimum
B = 0.0 V or GND

Outputs H > 1.5 V
L < 1.5 V

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6.3 Ordering data. The acquisition document should specifiy the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND - - - - - - - - - - - - - - Ground zero voltage potential.
I_{IN} - - - - - - - - - - - - - - Current flowing into an input terminal.
V_{IN} - - - - - - - - - - - - - - Voltage level at an input terminal.

6.5 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.6 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54F283

6.7 Manufacturers' designators. Manufacturers' circuits which form a part of this specification are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturers' designations.

Device type	A	B	C
	Fairchild Semiconductor	Motorola Inc.	Signetics Corp.
01	X		

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6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 17, 19, 85, 99
DLA - ES

User activities:
Army - SM

Agent:
DLA - ES

Preparing activity:
Air Force - 17

(Project 5962-F656)

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